Module 5 Properties, Postulates, and Theorems

Properties:

1. **Properties of a Circle:**

- A radius is half the length of a diameter.
- A chord is any line segment that connects two points on a circle.
- A diameter is the longest chord in a circle.
- A central angles is formed by two radii and its vertex is at the center of the circle.
- An inscribed angle is formed by two chords that share an endpoint, and its vertex is on the circle.
- A tangent is a line that touches the circle at exactly one point, the point of tangency.
- A secant is a line that intersects the circle at two points.
- The measure of a minor arch is less than 180 degrees, while a major arc is greater than 180 degrees.

2. Circle Properties:

- All radii in a circle are equal.
- A **diameter** is twice the length of a **radius**

3. Arc Measure Property: The measure of a minor arc is equal to the measure of its corresponding central angle. The measure of a major arc is $360°$ minus the measure of the minor arc.

4. Definition of an Inscribed Circle:

- A circle is inscribed in a triangle if it touches all three sides of the triangle at exactly one point each.
- The center of the inscribed circle is the incenter, which is found by constructing the angle bisectors of the triangle.

5. Definition of a Circumscribed Circle:

- A circle is circumscribed around a triangle if it passes through all three vertices of the triangle.
- The center of the circumscribed circle is the circumcenter, which is found by constructing the perpendicular bisectors of the triangle's sides.

Postulates:

1. **Inscribed Angles Intercepting the Same Arc:** If two inscribed angles intercept the same arc, they have equal measures.

2. **Arc Addition Postulate:** The measure of an arc formed by two adjacent arcs is the sum of the measures of the two arcs.

3. **Triangle Inscribed in a Circle:** If a triangle is inscribed in a circle and one of its sides is a diameter, then the triangle is a right triangle, and the angle opposite the diameter is $90°$.

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Theorems:

1. **Tangent Lines Theorem:** If a line is tangent to a circle, then it is perpendicular to the radius drawn to the point of tangency.

2. **Converse of Tangent Lines Theorem:** If a line is perpendicular to a radius of a circle at its endpoint on the circle, then the line is tangent to the circle.

3. **Theorem of Intersecting Chords:** If two chords intersect inside a circle, the product of the segments of one chord equals the product of the segments of the other chord $(\text{segment1}) \cdot (\text{segment2}) = (\text{segment3}) \cdot (\text{segment4})$.

4. **Diameter-Chord Perpendicularity Theorem:** If a diameter is perpendicular to a chord, then it bisects the chord into two equal parts.

5. **Diameter-Chord Perpendicularity Converse:** If one chord is a perpendicular bisector of another chord, then the first chord is a diameter.

6. **Two Tangents Theorem:** If two tangent segments are drawn to a circle from the same external point, they are congruent.

7. **Two Secants Theorem:** If two secant segments are drawn from an external point, the product of the external segment and the total length of one secant equals the product of the external segment and the total length of the other secant $(external_1)(total_1) = (external_2)(total_2)$.

8. **Secant-Tangent Theorem:** If a tangent and a secant are drawn from an external point, the square of the length of the tangent equals the product of the external segment and the total length of the secant $(tangent)^2 = (external)(total)$.

9. **Central Angle Theorem:** The measure of a central angle is equal to the measure of its intercepted arc.

10. **Inscribed Angle Theorem:** The measure of an inscribed angle is half the measure of its intercepted arc.

11. **Theorem of Angles Inside a Circle (Intersecting Chords Theorem):** If two chords intersect inside a circle, the measure of the angle formed is half the sum of the intercepted arcs.

$$m\angle = \frac{1}{2}(Arc1 + Arc2)$$

12. **Theorem of Angles Outside a Circle (Exterior Angles Theorem):** If two secants, two tangents, or a secant and a tangent intersect outside the circle, the angle formed is half the difference of the intercepted arcs.

$$m\angle = \frac{1}{2}(Larger\ Arc - Smaller\ Arc)$$

13. **Tangent-Chord Angle Theorem:** If a chord and a tangent meet at the point of tangency, the angle formed is half the measure of the intercepted arc.

$$m\angle = \frac{1}{2}(Intercept\ Arc)$$

14. **Cyclic Quadrilateral Theorem:** If a quadrilateral is inscribed in a circle, then its opposite angles are supplementary.